

Faculty Vitae: Betul Bilgin

1. Education

Ph.D. Chemical Engineering, Michigan State University, East Lansing, MI, 2014
M.S. Biomedical Engineering, University of Minnesota, Minneapolis, MN, 2008
B.S. Chemistry, Fatih University, Istanbul, Turkey, 2005

2. Academic Experience

- Department of Chemical Engineering, University of Illinois at Chicago, Clinical Professor, 2025-Present, Full-Time
- Department of Chemical Engineering, University of Illinois at Chicago, Clinical Associate Professor, 2021-2025, Full-Time
- Department of Chemical Engineering, University of Illinois at Chicago, Clinical Assistant Professor, 2015-2021, Full-Time
- Department of Chemical Engineering, University of Illinois at Chicago, Visiting Lecturer, 2014-2015, Part-Time

3. Non-Academic Experience

Not applicable

4. Certifications or professional registrations

- UIC Inclusive Education Scholars' Certificate Program (IESCP) Spring 2023
- UIC Equity Advocates Certificate Spring 2022
- Entrepreneurship Minded Learning (EML) June 16th 2019
- Tampa, FL
- Faculty Summer Institute May 20th -22nd 2019
Urbana-Champaign, IL
- BASF Faculty Workshop on Process Safety July 23rd-26th 2018
Wyandotte, MI
- ChemE Summer School July 29th-August 3rd 2017
NC State University, Raleigh, NC
- National Effective Teaching Institute – 1 May 21st-23rd 2017
Philadelphia, PA

5. Current membership in professional organizations

American Institute of Chemical Engineers (AIChE)
American Society for Engineering Education (ASEE)

6. Honors and awards

- UIC Teaching Scholar (Spring 2017)
- American Institute of Chemical Engineers (AIChE) 35 under 35 award in education category (August 2017)
- American Society for Engineering Education (ASEE) 20 under 40 award (June 2018)
- UIC Teaching Sustainability Initiative Spring (2020 and 2024)
- UIC Teaching Recognition Program (TRP) Award (Spring 2021)

- *UIC COE Harold Simon Award (Spring 2022)*
- *UIC COE Advising Award (Spring 2023)*
- *UIC COE Teaching Award (Spring 2022-24)*
- *ASEE Ray Fahien Award (June 2024)*

7. Service activities

- *AIChE Education Division 2nd Vice Chair* (November 2025-present)
- *ASEE Chemical Engineering Division Director* (July 2024-present)
- *AIChE Education Division Teaching Focus Faculty Leadership Board (Summer 2023-present)*
- *COE Summer Camp Chemical Engineering Advisor* (Summer 2022-2023-2024)
- *CHE Senior Design Showcase* (April 2022-2024)
- *AIChE Student Chapter Advisor* (January 2024-present)
- *Department Advisory Committee member* (September 2023-present)
- *Departmental Action Team (DAT) and CHEME Connect* (January 2023-present)
- *Summer CHEM-E Club and ACCESS Club- Founder/Chair* (May 2020-2022)
- *NSF S-STEM Project- An Integrated Program for Recruitment, Retention, and Graduation of Academically Talented Low-Income Engineering Students – co-PI* (September 2019-present)
- *COE Non-Tenure Track Faculty Committee-Member* (Fall 2018-present)
- *Instructional Delivery Planning (IDP) Committee – Member* (May-June 2020)
- *Diversity, Recruitment and Retention Committee-Chair* (Fall 2017-present)
- *Industry and Public Relation Committee- Chair* (Fall 2015-present)
- *Student and Alumni Development* (Spring 2015-present)
- *Search Committees*
 - *Clinical Faculty Hire Search Committee* (Summer 2020)
 - *Associate Dean of Undergraduate Affairs* (Summer 2020)

8. Selected Publications

- **Bilgin, Betul**, Christopher VH-H. Chen, and Stephanie Butler Velegol. "Generative AI in Chemical Engineering Education: Rebuilding Thermodynamics, Material and Energy Balances and Kinetics Courses with AI and Chemical Engineering Students' Perception of AI." In *2025 ASEE Annual Conference & Exposition*. 2025.
- Revelo, Renata A., Melissa Espindola, **Betul Bilgin**, and Houshang Darabi. "WIP Exploring the Interplay Between Social Identities and Engineering Identity Formation." In *2025 ASEE Annual Conference & Exposition*. 2025.

- **Bilgin, Betul.** "CHE CALCULATOR®: Empowering Students with Computational Tools for Real-World Engineering Challenges." In *2025 ASEE Annual Conference & Exposition*. 2025.
- **Bilgin, Betul,** Dania Nazimuddin, Louie Edano, and Nadia Nikolova. "Fostering Early Engineering Identity Through K-12 Outreach: Insights from a Three-Week Summer Program." In *2025 IEEE Global Engineering Education Conference (EDUCON)*, pp. 1-7. IEEE, 2025.
- **Betul Bilgin.** "Fostering Diversity, Equity, and Inclusion in Engineering Education- A Case Study of UIC Chemical Engineering Department" In *2024 ASEE Annual Conference & Exposition*, 2024.
- **Betul Bilgin,** Daisaq Cyrilla, Naomi Elisabeth Groza, Ruben Lopez. "Assessing Chemical Engineering Students' Perspectives on Data Science and Its Integration in the Academic Curriculum: Implications for Industry Readiness" In *2024 ASEE Annual Conference & Exposition*, 2024.
- **Betul Bilgin,** Hasiya Najmin Isa, Emily Seriruk, Cody Mischel, James Pellegrino. "Bridging the Gap between Industry and Academia, and Developing Students' Engineering Identity." In *2023 ASEE Annual Conference & Exposition*, 2023.
- Houshang Darabi, Rezvan Nazempour, Peter C Nelson, Renata A Revelo, Shanon Marie Reckinger, Didem Ozevin, **Betul Bilgin,** Anthony E Felder. "Board 375: Reporting the Progress and Performance Evaluation of an Ongoing Integrated Program for Recruitment, Retention, and Graduation of High-Achieving, Low-income Engineering Students." In *2023 ASEE Annual Conference & Exposition*, 2023.
- **Betul Bilgin,** Cody Mischel, Lewis Wedgewood, James Pellegrino, and Vikas Berry. "Impacts of implementing up-to-date industry problems on engineering identity development." In *2022 ASEE Annual Conference & Exposition*. 2022.
- Nazempour, Rezvan, Houshang Darabi, **Betul Bilgin,** Anthony Felder, Shanon Reckinger, Renata Revelo, Peter Nelson, Jeremiah Abiade, and Didem Ozevin. "An Integrated Program for Recruitment, Retention, and Graduation of Academically Talented Low-Income Engineering Students: Lessons Learned and Progress Report." In *2022 ASEE Annual Conference & Exposition*. 2022.
- **Betul Bilgin,** Cody Mischel, Lewis Wedgewood, James Pellegrino, and Vikas Berry. "Developing Professional Identity: Integrating Academic and Workplace Competencies within Engineering Programs." In *2022 ASEE Annual Conference & Exposition*. 2022.
- **Bilgin, Betul,** Anthony E. Felder, Houshang Darabi, Rezvan Nazempour, Shanon Reckinger, Renata A. Revelo, and Didem Ozevin. "Looking Ahead: Structure of an Industry Mentorship Program for Undergraduate Engineering Students." *Advances in Engineering Education* 10, no. 3 (2022).

9. Professional Development

I regularly attend and present at major engineering and education conferences, including the *ASEE Annual Conference*, *AIChE Annual Meeting*, *IEEE EDUCON*, and *AIChE Chicago Section* events, focusing on engineering education, sustainability, electrochemistry, and generative AI in teaching and learning.

I applied for *ABET Program Evaluator Training (PEV)* and regularly participate in UIC faculty development seminars and national workshops on course redesign, inclusive teaching, assessment, safety, instrumentation, and proposal development (NSF IUSE, RFE, REU). My

DEI-focused professional development includes participation in the certificate programs and ongoing leadership training as Chair of the departmental DEI Committee.

My professional growth is supported by active research collaborations across UIC and nationally, including work on engineering identity, data-driven learning, sustainability, and the *NSF ERC AMaZED* initiative. I also develop innovative pedagogical tools such as *AI-integrated curriculum*, CHE CALCULATOR®-based assignments, and K–12 outreach programs.

I regularly engage with industry partners through Senior Design mentorship coordination, strengthening my understanding of current engineering applications and industry needs.